Magnetosphere and changing fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use evidence to distinguish a magnetic-field model from an unsupported space-weather claim.

SUCCESS CRITERIA - I can...

1. Connect changing magnetic fields with induced electric effects.
2. Distinguish an assumption from an observation.
3. Name evidence required for a space-weather prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “magnetosphere” and “induction” to compare your model or result. Sentence starters are optional.

magnetosphere: a region strongly influenced by a body’s magnetic field

induction: producing an emf through a changing magnetic flux

flux: a measure of magnetic field passing through a chosen area

01 Warm-up: bring back an earlier idea

Near an ideal long straight wire, current becomes 3 times as large and distance becomes 5 times as large. **Compare** B.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

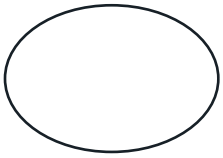
Reference: $|emf_{avg}| = N|\Delta\Phi|/\Delta t$; $\Phi = BA$.
Uniform perpendicular field for $\Phi = BA$. Use supplied/measured field changes; an asteroid alone is not evidence of an electromagnetic pulse.

Worked reasoning: A changing magnetic environment can induce voltages in conducting loops. A static magnetic field alone is not the same as a changing flux through a particular circuit.

Your turn: Sketch a conducting loop and field arrows; show one way flux can change.

Sentence starter: The … in my model represents … . Its label shows … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

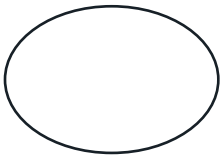
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw a coil in a changing magnetic field.
Label what changes and where an induced emf could appear.

Sentence starter: The … in my model represents … . Its label shows … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

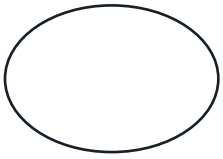
04 Apply the physics

A hypothetical 10-turn loop of area 5 m² experiences a uniform perpendicular field change of 2×10^{-6} T over 7 s.
Label the given quantities and calculate average emf magnitude. Does this calculation establish what caused the field change?

Word help: uniform = constant throughout the stated interval

Sentence starter: I found … [unit]. This value means … in this situation.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

05 Explain your reasoning

A headline claims an approaching rock will disable power grids through an EMP, but gives no field measurements or mechanism.
Name two pieces of evidence needed before accepting it.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Two series components have drops 4 V and 6 V. A charge of 8 C passes through each.
Find each energy transfer and the total.
Label voltage, charge and energy with their distinct units.

Word help: voltage = electric potential difference; energy transferred per charge

Sentence starter: I found … [unit]. This value means … in this situation.

Initial / supply _____ Final / component _____

V: ____ J/C V: ____ J/C

Charge: ____ C Energy transfer: ____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

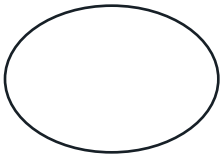
Sentence starters are optional. You may plan in any language.

07 Exit check

Explain why a one-loop calculation cannot by itself predict effects on a whole power grid.

Sentence starter: I predict … . My model supports this because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

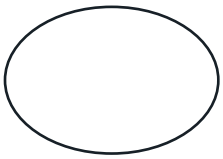
08 STRETCH: test the model

A one-turn 100 m² loop is perpendicular to a uniform field that changes by 2 microtesla in 0.5 s.
Estimate emf.
Explain why this toy result is not a full forecast for a power grid.

Word help: estimate = find an approximate value

Sentence starter: Under the assumption …, the evidence … supports … because … .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3